Stats And Data Science Sampling (1)

Week 1 Class Overview

- Introduction to modules and practicals.
- Initial confusion regarding practicals:
  - Two lecturers canceled practicals for personal reasons.
  - Apologies for any confusion caused.

Sampling Frame

- Concept introduction:
  - A sampling frame is a crucial tool in sampling that may not always exist or may lack accuracy/completeness.
- **Definition:** A sampling frame is defined as the set of items or individuals from which the sample is drawn.
- Example of a sampling frame:
  - Researching students at Stellenbosch University:
  - Contacting the registrar's office to access records of registered students.
- Importance of census:
  - Conducted every few years to track the population within a country.

Types of Sampling

Non-probability Sampling

- Brief overview to acknowledge existence, focusing on probability sampling instead.

Types of Non-probability Sampling

1. **Convenience Sampling:**
   - Sampling based on availability or ease of access.
   - Example: Surveying everyone passing on a street corner.
   - Implication:
     - May lead to bias, especially if the sample does not represent the larger population (e.g., surveying only BCom students).
2. **Judgment Sampling:**
   - Selection based on expert advice.

- Example: Consulting an anthropologist before studying a remote tribe.
- Note:
  - Involves human subjectivity; lacks random selection.

Probability Sampling

- Focus on ensuring an equal chance of selection, negating bias.

Types of Probability Sampling

1. **Simple Random Sampling:**
   - Each item has an equal chance of being selected.
   - Selection with or without replacement:
     - **With replacement:** An item is placed back into the population after being sampled, allowing potential re-selection.
     - **Without replacement:** An item is permanently removed from the population after being sampled; selection probabilities change after each draw.
   - Example with probabilities:
     - If sampling from 10 items:
     - First selection probability: $rac110$
     - Second selection probability after one item removed: $rac19$ and so forth.
2. **Systematic Sampling:**
   - Involves selecting every k-th item after a randomly selected start.
   - Notation:
     - Large N = population size, small n = sample size.
     - Example calculation:
     - If N = 40 and n = 4,
     - k = rac{N}{n} = rac{40}{4} = 10 .
   - Selecting a random number between 1 and k to begin the selection process provides systematic representation across the population.
   - Advantages:
     - More efficient for organized data, maintaining a good representation without clustering bias.
3. **Stratified Sampling:**
   - Ensures representation of specific characteristics or strata within a population.
   - Definition of strata:
     - Subgroups that may include characteristics like gender, age, or educational background.
   - Example:
     - If a population is 30% female, 70% male, targeting this ratio.

    - Sampling proportionately from each subgroup ensures accurate representation.

4. **Cluster Sampling:**
    - A pre-existing cluster or group within a population is selected.
    - Example:
        - Election districts in a city allow for sampling within small geographical areas.
    - Assumption: Clusters must be similar enough to represent the total population —this may introduce bias if not properly accounted for.

Comparison of Sampling Methods

- **Simple Random & Systematic Sampling: **
    - **Advantages:** Relatively easy to implement.
    - **Disadvantages:** Potential lack of necessary representation of population characteristics.
- **Stratified Sampling:**
    - **Advantages:** Ensures proportional representation of the population’s key characteristics.
- **Cluster Sampling:**
    - **Advantages:** Convenient, cost-effective but risks sacrificing accuracy.
    - **Disadvantages:** Need sufficient sample clusters to maintain accuracy.

Final Problem Scenario for Consideration

- Scenario description:
    - Population size (Large N): 500 invoices needing error checks.
    - Sample size (Small n): Random selection of 20 invoices to verify errors.
    - Aim: Determine error prevalence in chosen sample to assess overall quality.
    - Implication: Further action could be necessary based on the sample findings (e.g., larger sample checks if multiple errors are found).